IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

DIGITAL SIGNAL PROCESSING

Thursday, 8 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.T. Stathaki
Second Marker(s) : W. Dai

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1. (a) Consider the causal, complex coefficient transfer function shown below:

$$H_M(z) = \frac{\sum_{n=0}^{n=M-1} d^*[M-n]z^{-n} + z^{-M}}{1 + \sum_{n=1}^{n=M} d[n]z^{-n}}, M > 1 \quad (\mathbf{1})$$

(i) Show that the transfer function in **(1)** above is an allpass filter. **[5]**

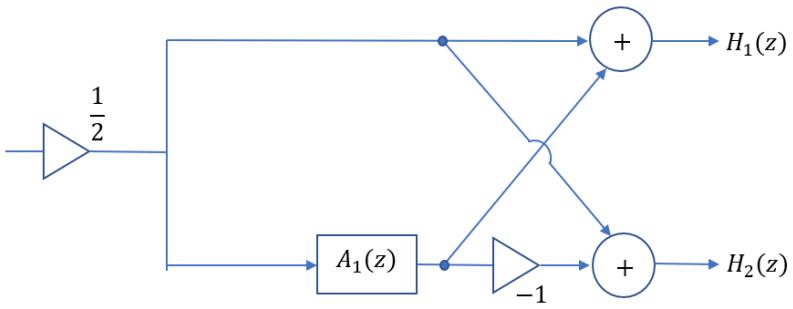
(ii) Comment on the locations of the zeros and the poles of a causal, stable, allpass filter's transfer function within the $z-$plane, with respect to the unit circle. Justify your answer. **[5]**

(b) Given an allpass filter

$$H(z) = \frac{d + ez + fz^2}{1 + bz + cz^2}$$

with poles at $\frac{1}{2}$ and $\frac{1}{3}$, find the coefficients b, c, d, e, f. **[5]**

(c) In the structure below, $A_1(z)$ is a first-order allpass transfer function. Find the type of filters $H_1(z)$ and $H_2(z)$ (lowpass etc.).



[5]

(d) Consider the cascade of two causal LTI systems with impulse responses $h_1[n] = a\delta[n] + b\delta[n-1]$ and $h_2[n] = c^n u[n]$, a, b, c are real coefficients and $|c| < 1$. Determine the frequency response $H(e^{j\omega})$ of the overall system. Find all the possible sets of values of a, b, c for which the overall transfer function has an amplitude equal to 1 for all frequencies. **[5]**

2. (a) Design a length-5 FIR bandpass filter with an antisymmetric impulse response $h[n]$, satisfying the following magnitude response values: $\left|H(e^{j0.3\pi})\right| = 0.3$ and $\left|H(e^{j0.6\pi})\right| = 0.8$. **[4]**

(b) We have shown that a real-coefficient FIR transfer function $H(z)$ with a symmetric impulse response has a linear phase response. As a result, the all-pole IIR transfer function $G(z) = \frac{1}{H(z)}$ will also have a linear-phase response. What are the practical difficulties in implementing $G(z)$? Justify your answer mathematically. **[4]**

(c) A simple averaging filter is defined by the following input-output relationship:

$$y[n] = \frac{1}{N}(x[n] + x[n-1] + \cdots + x[n-(N-1)])$$

(i) Let $N = 4$. Determine the filter's transfer function and its zeros and poles. **[3]**
(ii) Determine a general form for zeros and poles for any N. **[3]**
(iii) A recursive implementation of a filter refers to writing the current output as a function of the previous output and the current and past inputs. By comparing $y[n]$ and $y[n-1]$ determine a recursive implementation of the filter. **[3]**

[**Hint:** You may use the relationship $\sum_{i=0}^{N-1} z^{-i} = \frac{1-z^{-N}}{1-z^{-1}}, \ z \neq 1$.]

(d) An FIR digital filter has the transfer function

$$H(z) = (1 - z^{-1})^3(1 + z^{-1})^3$$

(i) Sketch the pole-zero diagram of this system. **[4]**
(ii) Sketch roughly the amplitude response of the above filter. Would you classify this as a low-pass, high-pass, band-pass, or band-stop filter? Please briefly explain. **[4]**

3. (a) (i) Explain using mathematical relations, where each subset of the z −plane is mapped on the s −plane through the bilinear transformation. **[5]**

(ii) Prove that the relationship between the continuous-time angular frequency and the discrete-time angular frequency is non-linear. **[5]**

(b) A given real-coefficient, digital IIR lowpass filter has a rational transfer function $H_L(z)$ and a cutoff frequency ω_c. The transfer function of $H_L(z)$ is transformed by replacing z by $F(z) = \frac{az+b}{cz+d}$ to a real highpass rational transfer function $H_H(z) = H_L(F(z))$ of the same order as $H_L(z)$ but different cutoff frequency. By using the constraints $H_H(e^{j\pi}) = H_L(e^{j0}) = 1$ and $H_H(e^{j0}) = H_L(e^{j\pi}) = 0$, derive relationships among the parameters a, b, c, d and explain what type of filter is $F(z)$. **[5]**

(c) Consider the Finite Impulse Response (FIR) filter transfer functions $G_1(z) = \frac{1}{4}(z + 2 + z^{-1})$ and $G_2(z) = \frac{1}{4}(-z + 2 - z^{-1})$. Explain the type of filters $G_1(z)$ and $G_2(z)$ (lowpass etc), by using the following two approaches:

(i) Experimental approach. In this approach you will apply the filters to the signal $x[n] = u[n]$, with $u[n]$ the discrete unit step function and observe the effects they have on $x[n]$. **[5]**

(ii) Mathematical approach. In this approach you must find the frequency response of the two filters and, based on that, find the type of the filters. **[5]**

4. (a) Consider the multirate structure of **Figure 1** below, where $H_0(z), H_1(z)$ and $H_2(z)$ are ideal, zero phase, real-coefficient lowpass, bandpass and highpass filters respectively, with frequency responses as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{2\pi}{3})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \frac{2\pi}{3})$$

$$H_2(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \pi)$$

The function $u(\omega)$ is the well-known unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (-\frac{9\omega^2}{\pi^2} + 1)\left(u(\omega) - u\left(\omega - \frac{\pi}{3}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n]$ and $y_2[n]$. Notice that all frequency domain representations above are provided only within the range $[0, \pi]$.

[15]

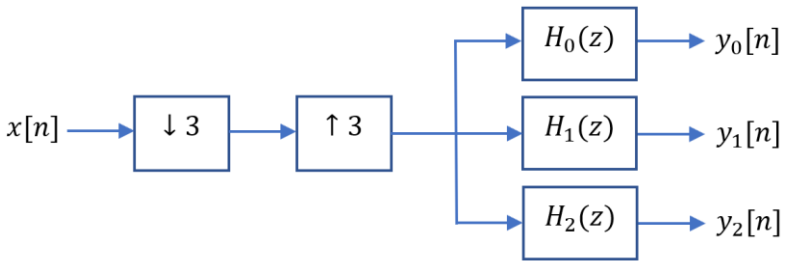


Figure 1

(b) Develop an expression for the output $y[n]$ as a function of the input $x[n]$ for the multirate structure of **Figure 2** below.

[10]

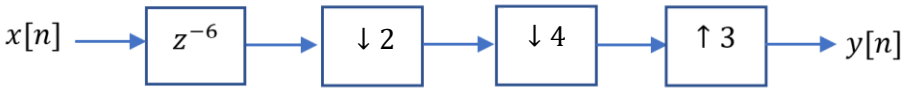


Figure 2